Student Name	ANSWERS			
Teacher	GSL	HAL	LLE	MDI

Chemistry Unit 3 SAC 3

Analysis of Second hand Data Report (Concepts from AOS 2 - Ch 7.4 to 14)

Friday 5 June, 2015

Reading time: 2:05pm to 2:10pm (5 minutes)
Writing time: 2:10pm to 3:15 pm (65 minutes)

QUESTION AND ANSWER BOOK
Structure of book

Section	*Number of questions*	*Number of questions to be answered*	*Number of marks*
A	12	12	12
B	5	5	49
			Total 61

- Students are permitted to bring into the assessment session: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.
- Students are NOT permitted to bring into the assessment session: blank sheets of paper and/or white out liquid/tape.

Materials supplied
- Question and answer book of 18 pages, including answer sheet for multiple choice questions on pg 7.
- A data booklet (Please do not write in the data booklet)

Instructions
- Write your **name** and circle your **teacher code** in the space provided above on this page.
- All written responses must be in English.
- Do NOT write on the data book.

At the end of the assessment session
- Wait silently for the supervisor to collect your **question and answer booklet**, and your **data book**.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the assessment session room.
Students must not disclose the contents of the task; to do so will be a breach of VCE guidelines and will be dealt with according to VCAA regulations.

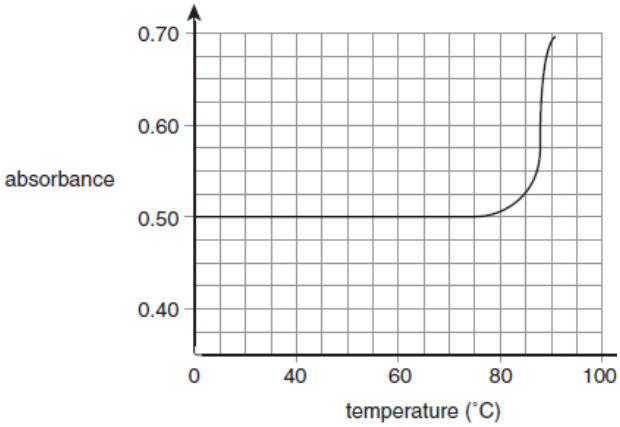
SECTION A: Multiple-choice questions

Instructions for Section A

Answer **all** questions in pencil on the answer sheet provided for multiple choice questions.
Choose the response that is **correct** or that **best answers** the question.
A correct answer scores 1, an incorrect answer scores 0.
Marks will **not** be deducted for incorrect answers.
No marks will be given if more than one answer is completed for any question.

Use the following information to answer Questions 1–3.

When double-stranded DNA is heated slowly the individual strands separate from each other over a narrow temperature range. The midpoint temperature (*T*m) of this separation is known as the melting temperature. During this transition from double helix to random coil arrangement, the absorbance at 260 nm of a particular DNA sample increases as shown in the graph below.



Question 1

What is the melting temperature of the DNA sample?

A. 50°C
B. 73°C
C. 87°C
D. 92°C

The melting temperature occurs halfway through the separation process, which starts at about 75°C and ends at about 100°C (by extrapolation). Halfway is approximately 87°C.
Answer is C

Question 2

Which level of structure of DNA, if any, would be disrupted during heating to determine the melting temperature of a DNA sample?

A. Primary only
B. Secondary only
C. Both primary and secondary
D. Neither primary nor secondary

The primary structure of DNA involves the covalent bonding of components to make a single strand. The secondary structure involves the interactions between complementary bases on separate strands. This occurs by hydrogen bonding. At the temperatures which cause separation of the DNA strands, hydrogen bonding will be disrupted, but covalent bonds will not.
Answer is B

Question 3

Which of the following nitrogenous bases in the DNA sample contribute most to a high melting temperature value?

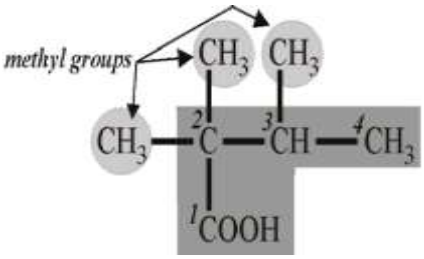
A. adenine and guanine
B. thymine and cytosine
C. adenine and thymine
D. cytosine and guanine

Adenine and thymine are bonded by two hydrogen bonds, whereas guanine and cytosine are bonded by three hydrogen bonds. A greater number of hydrogen bonds between the DNA strands will cause a higher melting temperature of a DNA sample. Answer is D

Question 4

The systematic name for the compound whose structure is shown below is

A. 2,3-dimethylbutanoic acid.
B. 2-methyl-2-propylpropanoic acid.
C. 2,2,3-trimethylbutanoic acid.
D. 2,2,3-trimethylpropanoic acid.



The structure contains a carboxy functional group, -COOH, **-oic acid.** Locate the longest carbon-carbon atom chain that contains the carboxy functional group. This has four carbon atoms which are joined by single carbon-carbon bonds, **Butanoic acid**
There are three methyl groups attached to the backbone. Two methyl groups attached to carbon atom numbered 2, and another methyl group attached to carbon atom numbered 3, **2,2,3-trimethylbutanoic acid.**

Question 5

The following are **incomplete** and **unbalanced** equations representing three types of chemical reactions that involve glucose. In reactions 1 and 3, product A is the same compound. In reactions 2 and 3, product B is the same compound.

reaction 1 $C_6H_{12}O_{6(aq)} \rightarrow C_2H_5OH_{(aq)}$ + product A
reaction 2 $C_6H_{12}O_{6(aq)} \rightarrow C_{12}H_{22}O_{11(aq)}$ + product B
reaction 3 $C_6H_{12}O_{6(aq)} \rightarrow$ product A + product B

Which one of the following correctly names reaction 3 and identifies product A and product B?

	Reaction 3	**Product A**	**Product B**
A.	Combustion	Water	Carbon dioxide
B.	Combustion	Carbon dioxide	Water
C.	Fermentation	Water	Carbon dioxide
D.	Fermentation	Carbon dioxide	Water

Reaction 1 represented fermentation and product **A** was **CO_2**. Reaction 2 represented a condensation reaction in which a disaccharide is produced from a monosaccharide. Product **B** was **H_2O**. Since one reactant in reaction C was $C_6H_{12}O_6$(aq), and the products were CO_2 and H_2O, it is perhaps most readily recognised as respiration or combustion **:** $C_6H_{12}O_6(aq) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$. Answer is B.

Question 6

A polypeptide has amino acids connected in the following ratio:

4 Glycine	:	2 Alanine	:	6 Valine
76.06 g mol^{-1}	:	89.09 g mol^{-1}	:	117.15 g mol^{-1}

The molar mass of the peptide would be closest to:

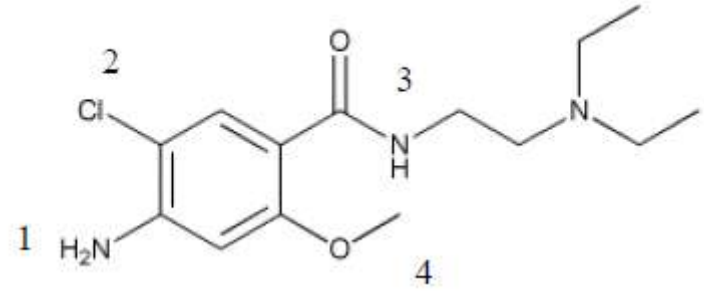
A. 951.3 g mol^{-1}
B. 969.3 g mol^{-1}
C. 987.3 g mol^{-1}
D. 1185.3 g mol^{-1}

M(peptide) = sum of M(amino acids) – m(H_2O)lost
To link twelve A.A.'s together, 11 molecules of water would be lost

Molar mass (peptide)
= ((4x 76.06) + (2 x 89.09) + (6 x 117.15)) – (11 x 18)
= 987.3 $gmol^{-1}$ Answer is C

Question 7

The molecule drawn below is known as metoclopramide. It is used to treat nausea and vomiting and is often used in conjunction with paracetamol.



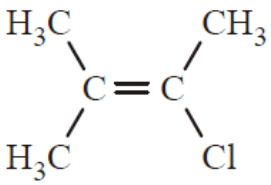
The four functional groups numbered on this molecule are, in order,

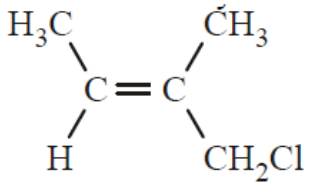
A. Amine, chloro, amide and ester
B. Amide, chloro, amine and ether
C. Amine, chloro, amide and ether
D. Amide, chloro, amine and ester

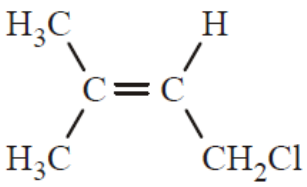
NH_2 is an amine; Cl obviously chloro or halo; CO- NH- is amide; C—O –C is ether
Answer is C

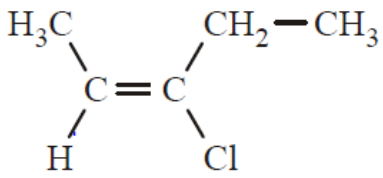
Question 8

Which one of the structures shown below is that for 1-chloro-2-methylbut-2-ene?

A.


B.


C.


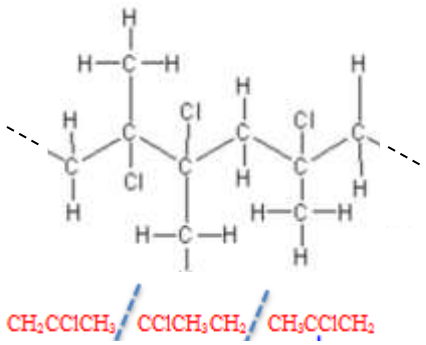
D.


Answer = B

The other structures are for:
A. 2-chloro-3-methyl-but-2-ene
C. 1-chloro-3-methyl-but-2-ene
D. 3-chloro-pent-2-ene

Question 9

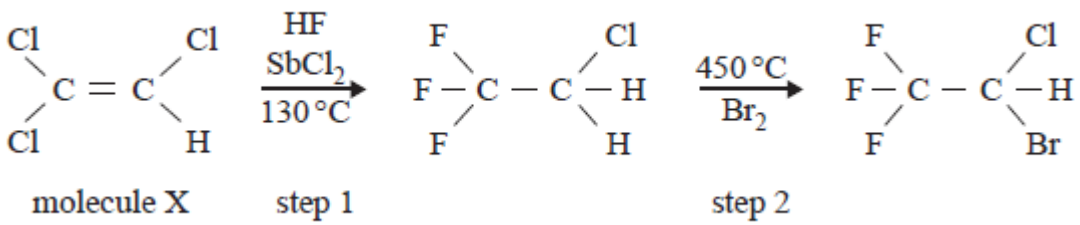
Below is a segment of a polymer chain. Choose the monomer (s) which makes up this polymer.



A. $C(CH_3)ClC(CH_3)Cl$ and $C(CH_3)ClCH_2$
B. $C(CH_3)ClCH_2$ only
C. CH_2CH_2 and $C(CH_3)ClCH_2$
D. $C(CH_3)ClC(CH_3)Cl$ and CH_2CH_2

Question 10

Halothane is a general anaesthetic. The following diagram represents the reaction pathway that produces halothane.



Which one of the following correctly identifies the type of reaction occurring in step 2 and correctly states the systematic name of molecule X?

	Type of reaction in step 2	Systematic name of molecule X
A.	substitution	1,1,2-trichloroethene
B.	addition	1,2,2-trichloroethene
C.	addition	1,1,2-trichloroethene
D.	substitution	1,2,2-trichloroethane

X-1,1,2-trichloroethene
Step 2 is a substitution reaction, H is substituted by Br; Answer is A

Question 11

Which of the following statements regarding biofuels is **incorrect**?

A. The use of biofuels increases the overall amount of CO_2 in the atmosphere when combusted.
B. The reaction that produces some biofuels is similar to the formation of triglycerides in the body in that ester linkages are created during the reaction.
C. Heat, acids and bases can be used to accelerate the rate of formation of biofuels.
D. The type of alkyl ester that is formed during the production of biofuel is determined by the type of alcohol used as a reactant.

Even though biofuels undergo combustion and release CO_2, they also consumCO_2 during their growth. Therefore they do not increase the amount of CO_2 in the atmosphere (they are carbon neutral fuels). Answer = A

Question 12

Stearic acid ($C_{18}H_{36}O_2$), oleic acid ($C_{18}H_{34}O_2$) and linoleic acid ($C_{18}H_{32}O_2$) are all fatty acids that can form triglycerides. When comparing the three acids it can be said that:

A. Stearic acid is the most saturated fatty acid and has the lowest melting point.
B. The order of melting points and degree of unsaturation increases in the following order: linoleic acid, oleic acid, stearic acid.
C. Oleic acid and linoleic acid are both polyunsaturated fatty acids.
D. Stearic acid is the only fatty acid that will not react with Br_2, and has the highest melting point.

Stearic acid is the only saturated acid in the list ($C_nO_{2n}O_2$). It will not react with Br_2 since it has no double bonds. It will have the highest melting point due to the regular structure of the hydrocarbon chain which allows for stronger dispersion forces to exist between its molecules compared to the other two acids.
Answer is D

End of Section A

Section A Multiple Choice Answer Sheet

This answer sheet must not be detached from this booklet.

INSTRUCTIONS FOR MULTIPLE CHOICE ANSWER SHEET

Use a **PENCIL** for **ALL** entries. For each question, shade the box which indicates your answer.
Marks will **NOT** be deducted for incorrect answers.
NO MARK will be given if more than **one** answer is completed for any question.
If you make a mistake, **ERASE** the incorrect answer - **DO NOT** cross it out.

1	A	B	**C**	D
2	A	**B**	C	D
3	A	B	C	**D**
4	A	B	**C**	D
5	A	**B**	C	D
6	A	B	**C**	D
7	A	B	**C**	D
8	A	**B**	C	D
9	A	**B**	C	D
10	**A**	B	C	D
11	**A**	B	C	D
12	A	B	C	**D**

SECTION B

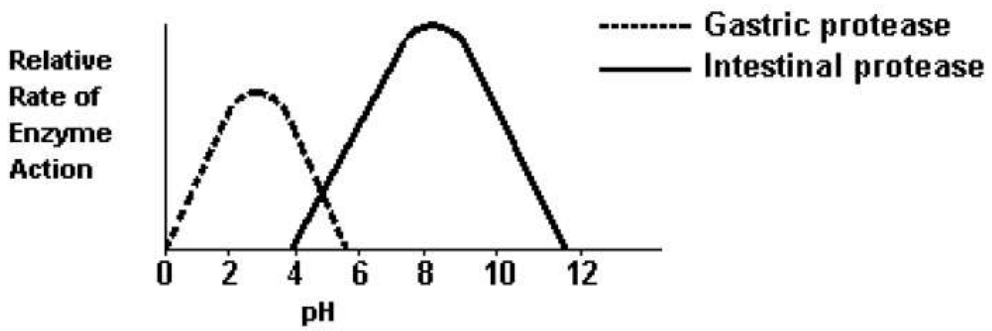
Instructions for Section B

Answer **all** questions in the spaces provided. Write using black or blue pen.
To obtain full marks for your responses you should:

- Give simplified answers with an appropriate number of significant figures to all numerical questions; unsimplified answers will not be given full marks.
- Show all working in your answers to numerical questions. No marks will be given for an incorrect answer unless it is accompanied by details of the working.
- Make sure chemical equations are balanced and that the formulas for individual substances include an indication of state; for example, $H_{2(g)}$; $NaCl_{(s)}$

Question 1 (11 marks)

The following graph shows the relative rate of action of two enzymes, Gastric protease and Intestinal protease versus pH.



a. Explain why a particular enzyme such as Gastric protease or Intestinal protease can only catalyse one specific reaction.
The 3-dimensional shape or tertiary structure of the active site must complement the structure/shape of the substrate molecules.

1 mark

b. i. Use the information in the graph to determine which of the following statements provides the best possible answer when both enzymes and their substrates are mixed in a single test tube. Give an explanation why you chose this answer and not the other three options.

I Only gastric protease would be active if the pH of the mixture was basic
II Intestinal protease would be more active than gastric protease at pH 4.
III Both enzymes would exhibit some activity at pH 5.
IV Gastric protease would be more active that intestinal protease at pH 6

III - Both enzymes would exhibit some activity at pH 5. [1 mark]

Gastric protease has no activity at pH 6 and Intestinal protease does not work at pH 4. Hence II and IV are incorrect. Gastric protease is active between pH 0 to 5, so option I is not correct. [1 mark]

*The second mark is awarded if students have used the graph data as evidence

ii. Which of the following statements best describes the relationship between pH and enzyme action? Give an explanation why you chose this answer and not the other three options.

I All enzymes work best at a neutral pH.
II Adding more acid does not affect the rate of activity of an enzyme.
III Enzymes function only in a pH range of 4.0 to 5.5.
IV The activity of an enzyme is affected by pH.

IV - The activity of an enzyme is affected by pH. [1 mark]

Gastric protease has an optimum pH of 3 and Intestinal protease has an optimum pH of 8, hence options I, II and III are incorrect. [1 mark]

*The second mark is awarded if students have used the graph data as evidence

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2 + 2 = 4 marks

c. A VCE student made a claim that **all catalysts** are solid. Explain why you agree or disagree with this claim, using a specific example that you studied this year.
The statement is false. Catalysts can be solids, liquids, gases or aqueous solutions; for example, concentrated sulfuric acid in esterification reactions. Solid H_3PO_4 is the catalyst used in addition of water to alkenes. Ultraviolet light is used in substitution/halogenation of alkanes. Enzymes are proteins (solids) which catalyse specific biological reactions. (1 mark for an example).

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1 mark

d. A VCE student investigated the effect of enzyme activity on the hydrolysis reaction of proteins at different temperatures. The student's report is provided below.

Background

Different enzymes, such as Gastric protease and Intestinal protease will increase the rate of hydrolysis reaction of proteins as the temperature increases.

Purpose

This experiment investigated the effect of using different enzymes on hydrolysis reactions of proteins at different temperatures.

Procedure

The amount of amino acids (products) was measured over time when Gastric protease was added to a volume of protein solution in a beaker at 20° C. The procedure was repeated using Intestinal protease.

Results

	Trial 1	**Trial 2**
Volume of Protein solution	50 mL	100 mL
Concentration of Protein solution	1.0 M	2.0 M
Enzyme	10.0 g crushed Gastric Protease	0.50 mol (solid piece) Intestinal Protease
pH of solution	3	3
Amount of amino acids produced in 10 seconds	15.0 g	0.2

Conclusion
Gastric protease produces greater amounts of amino acids at 20° C. It is a more effective enzyme than Intestinal protease.

The student's conclusion is not valid because the experimental design is flawed.

Critically review the student's experimental design. In your response, you should:

- identify and explain **three** improvements or modifications that you would make to improve the experimental design; and
- discuss the experimental outcomes you would expect regarding the effect of temperature on enzyme activity. Justify your expectations in terms of chemical ideas you have studied this year.

Improvements to the experiment:
In order to determine the effect of temperature on enzyme activity, specific variables need to be kept constant:

- Volume of protein solution
- Concentration of protein solution.
- Amount of Enzyme.
- Surface area of enzyme
- Using same insulation and same amount of insulation around the beaker.

The temperature of the reaction was conducted only at 20°C, so a conclusion cannot be made. Experiments should be conducted at body temperature (37°C) and higher (60°C) to ascertain how well each enzyme works.

The amount of amino acids produced should be measured at more time intervals, to ascertain whether there was no effect of enzyme activity using the Intestinal protease.

Only one trial was conducted with each enzyme. The results can be validated by repeating the experiment many times (more than 20 times at least).

The pH of the solution should not have been kept constant. A pH of 8 should be used for Intestinal protease trial.

Explanations:
It is difficult to make a conclusion without factoring in the optimum pH of each enzyme. Perhaps the enzymes chosen should have the same optimum pH or the pH should have been adjusted to allow for maximum activity for each enzyme. (1 mark)

Outside the optimal pH, Intestinal protease is denatured because the change in pH destroys the hydrogen bonding/dipole-dipole interactions and hence changes the 3-D shape of the active site of the enzyme. The enzyme can no longer bind to the substrate molecule and no activity should be observed.(1 mark)

The activity of the enzyme will decrease if the temperature decreases as there are a slower rate of fruitful collisions between the substrate molecule and the enzyme, whereas increasing the temperature (up to body temperature) increases the number of fruitful collisions and hence activity (1 mark) Temperatures higher than 37°C destroys the hydrogen bonding and affects the active site of the enzyme.

3 marks for identifying 3 improvements
2 marks for any 2 explanations. Or 1 mark for a discussion why controlled variables are used and 1 mark for a detailed explanation

5 marks

Question 2 (9 marks)

Over time, aspirin decomposes and its percentage purity decreases. To determine the percentage purity of a sample of aspirin, a VCE student followed the procedure outlined below.

A 0.2000g sample of aspirin was crushed and dissolved in 20.00mL of 0.2520 M sodium hydroxide. The equation for the reaction is:

$$C_9H_8O_{4(s)} + 2NaOH\ (aq) \rightarrow CH_3COONa\ (aq) + HOC_6H_4COONa\ (aq) + H_2O\ (l)$$

The solution was made up to 50.00mL and heated for 15 minutes. After heating, the volume was adjusted back to 50.00mL to compensate for any loss of solution during the heating process. 25.00mL of the solution was then titrated with 0.1056M HCl solution to find out how much NaOH was in excess. An average titre of 13.75mL was required to reach the end point.

a. i. Determine the amount, in mole, of HCl used in the titration.

n(HCl) = 0.01375 x 0.1056 = 1.452 x 10^{-3} mol [1 mark]

ii. Determine the amount, in mole, of NaOH left over in the 50.00mL solution.

$n(NaOH)_{inxs\ 25.00ml}$ = 1.452 x 10^{-3} mol

$n(NaOH)_{inxs\ 50.00ml}$ = 2 x 1.452 x 10^{-3} mol
= 2.904 x 10^{-3} mol [1 mark]

iii. Determine the amount, in mole, of NaOH that reacted with the aspirin.

n(NaOH)initially = 0.02000 x 0.2520 = 5.040 x 10^{-3} mol [1 mark]

n(NaOH)left over = 5.040 x 10^{-3} - 2.904 x 10^{-3} = 2.136 x 10^{-3} mol [1 mark]

1 + 1 + 2 = 4 marks

b. Determine the percentage purity of the aspirin sample.

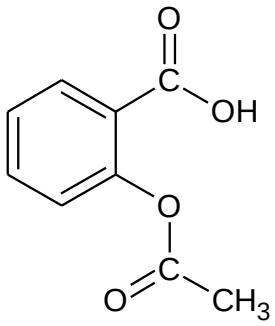
n(asprin) = 2.136 x 10^{-3}/2 = 1.068 x 10^{-3} mol [1 mark]

m(asprin) = 1.068 x 10^{-3} x 180 = 0.19224 g [1 mark]

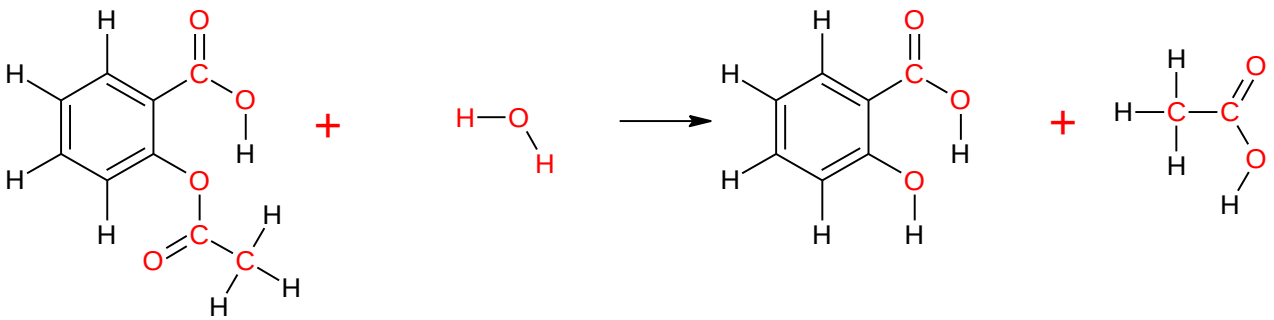
% purity = 0.19224/0.2000 x 100 = 96.12% [1 mark]

3 marks

c. The structure of aspirin is given below.



The active ingredient in aspirin is salicylic acid which is released after hydrolysis in the small intestine before moving into the blood stream.
Write a balanced equation showing the hydrolysis of aspirin to produce salicylic acid. Clearly **show all structures** (with bonds between atoms) for all reactants and products. A benzene ring may be used.



1 mark for correct structures of both reactants
1 mark for correct structures of both products.
Deduct 1 mark if structures have not been done e.g H_2O is a common mistake.

2 marks

Question 3 (11 marks)

A student was given four chemical substances labelled W, X, Y and Z and was told that they were all straight chain molecules with the following chemical formulae – **not in correct order**:

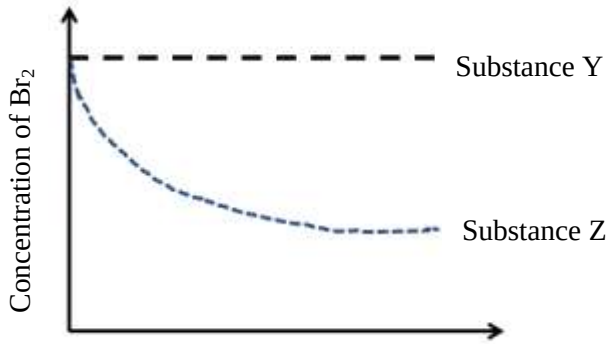
C_4H_{10} **C_4H_8** **$C_4H_{10}O$** **$C_4H_8O_2$**

To determine the identity of each substance from the list of molecules given, the student performed a number of experiments which are described below.

Experiment 1: Sodium carbonate solution was added to separate samples of Substance X and Substance W

	Observations
Substance X	No bubbles of gas observed.
Substance W	Bubbles of gas observed

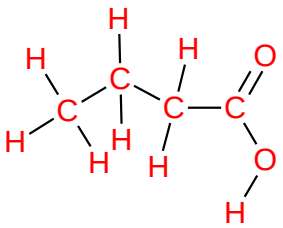
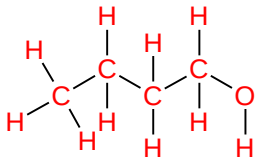
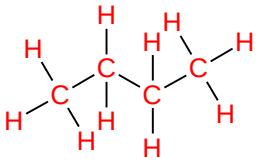
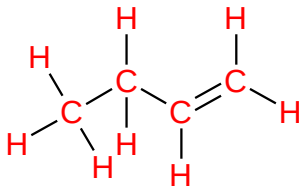
Experiment 2: Br_2 (aq) is added to a sample of Substance Y, and separately, to a sample of Substance Z. The change in Br_2 (aq) concentration was monitored and the following graph was obtained.



Experiment 3: A sample of Substance X and a sample of Substance Y were added to two separate test tubes containing deionised water.

	Observations
Substance X	Slightly soluble
Substance Y	Not soluble

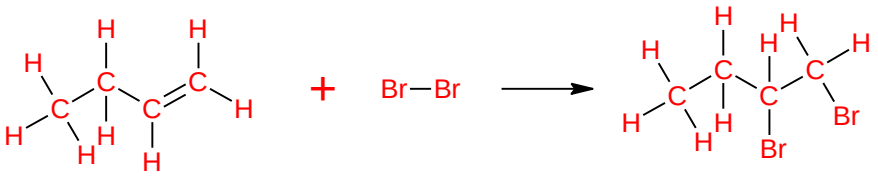
a. Use the information above to draw possible structures for Samples W, X Y and Z.

Substance **W** 	Substance **X** 
Substance **Y** 	Substance **Z** 

4 marks

b. Explain why the concentration of bromine decreases when it is reacted with Substance Z. In your answer, write a balanced equation to describe the reaction occurring.

The double bonds in unsaturated hydrocarbon molecules are broken and two atoms of bromine are added to the hydrocarbon via an addition reaction. In the process, the concentration of bromine decreases. (1 mark)



[1 mark]

Or

$CH_3CH_2CHCH_2 + Br_2 \longrightarrow CH_3CH_2CHBrCH_2Br$

2 marks

c. What other conditions would be required in order to observe a decrease in the concentration of bromine when mixed with Sample Z?

UV light [1 mark awarded for each student]

1 mark

d. Using *semi-structural formulae* show a possible reaction sequence that could be used to synthesise molecule X from an **unsaturated** compound.

$$CH_3CH_2CHCH_2 + H_2O \xrightarrow{H_3PO_4 \text{ catalyst}} CH_3CH_2CH_2CH_2OH$$

1 mark for semistructural formula
1 mark for catalyst.
Consequential marking applies from question 1a.

2 marks

e. Explain why alkanes and their corresponding alkenes have similar physical properties, but very different chemical properties.

Difference in reactivity: Due to the presence of the double bonds in alkenes. (1 mark)
Similarity in physical properties one of the following: Due to similar chain length, similar hydrocarbon composition or non polarity. (1 mark)

2 marks

Question 4 (10 marks)

a. The iodine number of a fatty acid is defined as the number of grams of iodine I_2 that reacts with 100 g of the fatty acid in an addition reaction.

i. Use this value (100 g) to determine iodine number of **linoleic acid**. Show all workings out.

linoleic acid is unsaturated and each molecule contains 2 C=C double bonds. One I_2 molecule reacts across each double bond.

$n(C_{18}H_{32}O_2)$ = 100 g / 280 g mol^{-1} = 0.357 mol (1 mark)

$n(I_2)$ reacting = 2 x $n(C_{18}H_{32}O_2)$ = 3 x 0.357 = 0.714 mol (1 mark)

$m(I_2)$ required = 0.714 mol x 254 g mol^{-1} = 181 g

Iodine number of linoleic acid is 181 (1 mark) [3 significant figures]

ii. Why does **arachidic acid** have an iodine number of zero?

Arachidic acid, $C_{19}H_{39}COOH$, has the molecular formula $C_{20}H_{40}O_2$ and so is saturated, i.e. has no C=C double bonds and does not undergo an addition reaction with iodine. (1 mark)

iii. Write a fully balanced equation showing the complete combustion of **arachidic acid**. Include states.

iv. $C_{20}H_{40}O_{2(l\ or\ s)} + 29O_{2(g)} \rightarrow 20CO_{2(g)} + 20H_2O_{(l)}$ (1 mark and 1 mark for states)

3 + 1 + 2 = 6 marks

b. A triglyceride consists of **three** omega-6 fatty acid molecules ($C_{34}H_{54}O_2$)

i. How many carbon to carbon double bonds (C=C) does omega-6 fatty acid molecules have?

7 C = C bonds

ii. What is the molecular formula of the triglyceride?

$C_{105}H_{164}O_6$

1+1 = 2 marks

c. Biodiesel is manufactured from vegetable oils. This involves the reaction of the oil with an alcohol in the presence of a catalyst in a two-step process. Firstly the oil molecules are broken down into their fatty acids and one 'other product'. The fatty acids then react with the alcohol to produce biodiesel.

i. Give the name and molecular formula of the 'other product' as mentioned in the first step.

Glycerol, $C_3H_8O_3$ (1 mark)

ii. Write a balanced equation for the production of the biodiesel methyl palmitoleate from its fatty acid.

The formula for palmitoleic acid is obtained from the data booklet (Table 9).

$C_{15}H_{29}COOH(l) + CH_3OH(l) \rightarrow C_{15}H_{39}COOCH_3(l) + H_2O(l)$ (1 mark for balanced equation)

1 + 1 = 2 marks

Question 5 (8 marks)

A small organic molecule has the molecular formula of the form $C_xH_yO_2$
The mass spectrum, infrared spectrum, ^1H NMR spectrum and ^{13}C NMR spectrum of this compound are provided on pages 17 and 18

a. On the infrared spectrum, label the peak (indicating the range) that corresponds to the presence of the functional group in this compound.
1 mark for labelling the **range** of the OH (alcohol) peak.
No marks for pointing to peak.

1 mark

b. Use the data provided to determine the values of x and y in $C_xH_yO_2$

x = 3 y = 8

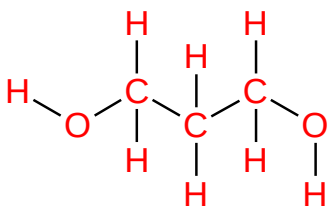
The molecular mass of $C_xH_yO_2$ is 76. (mass spectrum)
The compound has an OH (alcohol) absorption in the IR. Since the compound has two oxygen atoms, there are two hydroxyl groups present (Molar mass = 34). The molar mass of the remaining compound is 42. This suggests that there is no more than 3 Carbon atoms (M = 36) and hence there are 8 hydrogen atoms. (42 minus 36 =8 remaining)
x = 3 (1 mark) and y= 8 (1 mark)

2 marks

c. What specific information about the structure of the compound is provided by all the *splitting patterns* in the ^1H NMR spectrum?
A quintet suggests that there are two CH_2 groups in the same proton environment;
a triplet suggests that a neighbouring C atom has two hydrogen atoms in a different proton environment to the two CH_2 groups
and a singlet suggests that the hydrogen atoms are directly attached to the oxygen atoms (-OH groups)
1 mark for each splitting pattern explained.

3 marks

d. Draw the complete molecular structure for this molecule. Show all bonds.

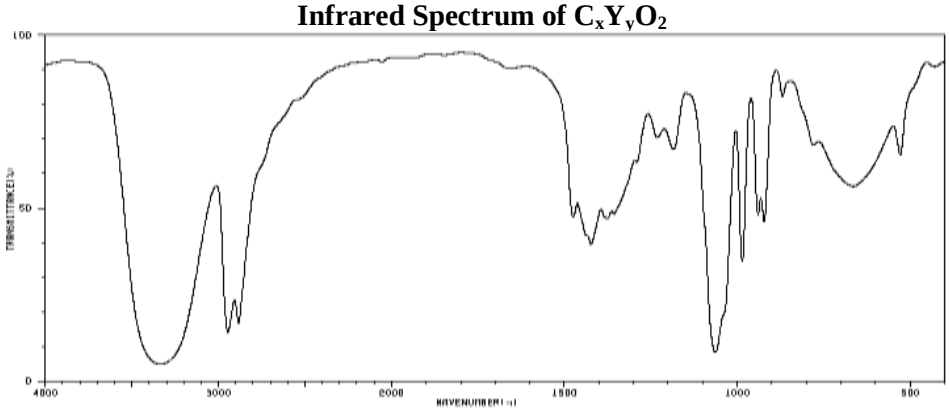


1 mark

e. Name the compound.
1,3-dihydroxy propane or propane-1,3-diol (1 mark)

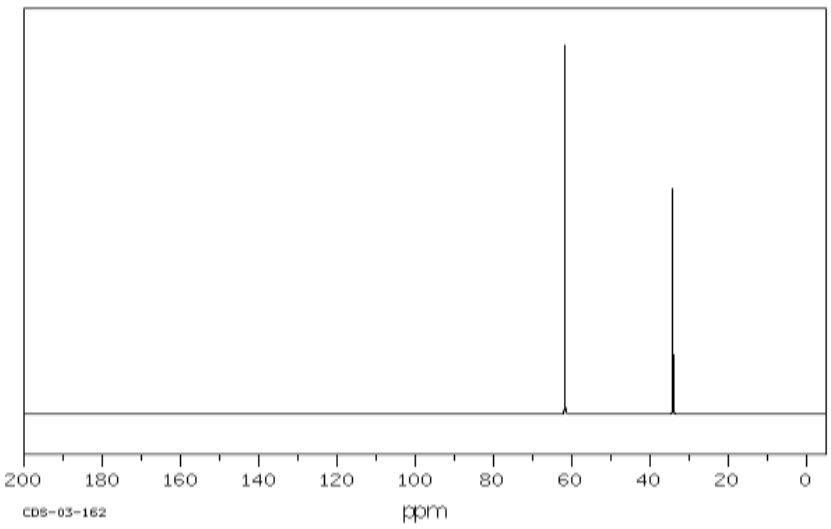
1 mark

Infrared Spectrum of $C_xY_yO_2$



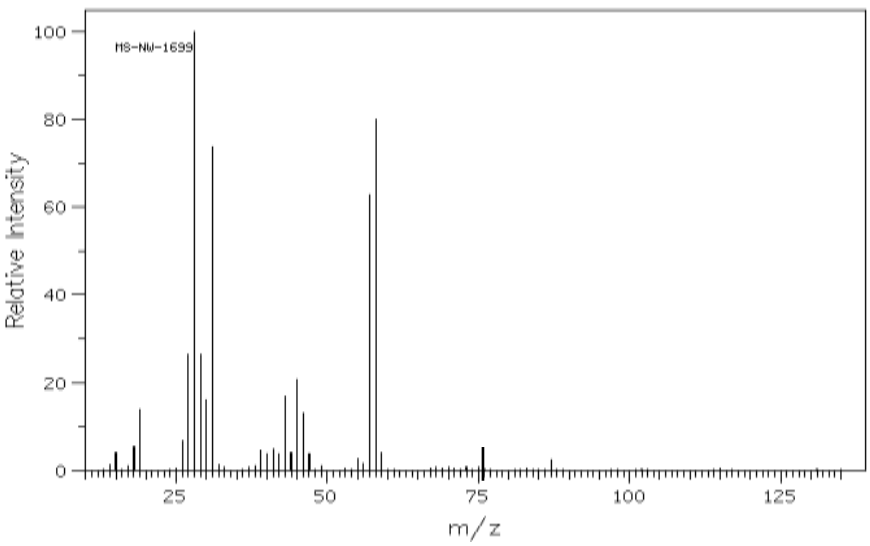
Data: National Institute of Advanced Industrial Science and Technology

^{13}C NMR spectrum of $C_xY_yO_2$



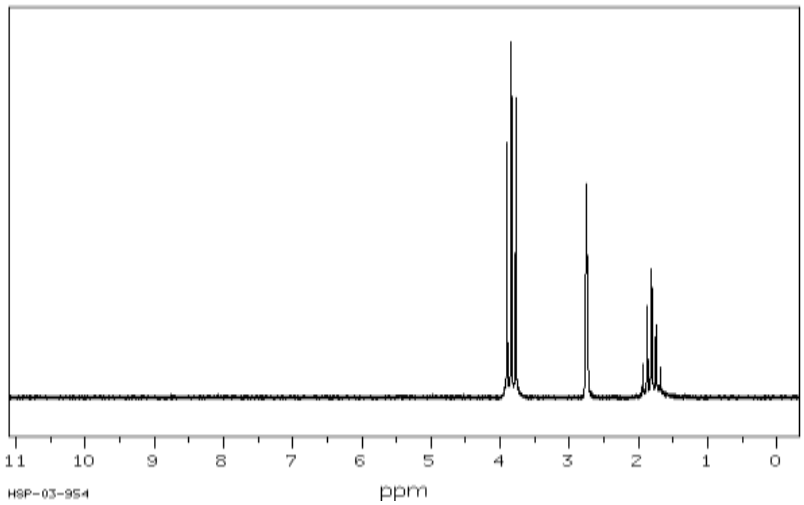
Data: National Institute of Advanced Industrial Science and Technology

Mass Spectrum of $C_xY_yO_2$



Data: National Institute of Advanced Industrial Science and Technology

^{1}H NMR Spectrum of $C_xY_yO_2$



Data: National Institute of Advanced Industrial Science and Technology

^{1}H NMR data

Chemical shift (ppm)	Peak splitting	Relative peak area
1.81	Quintet	1
2.75	Singlet	1
3.83	Triplet	2

Data: National Institute of Advanced Industrial Science and Technology